

## AACHEN, Germany

WMO: 105010

Lat: 50.783N

Lon: 6.095E

Elev: 663

StdP: 14.35

Time Zone: 1.00 (EUC)

Period: 82-03

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 2             | 13.8       | 19.2 | 6.5                           | 7.9 | 18.3 | 10.9 | 9.8 | 21.7 | 29.2                  | 48.3 | 26.2 | 46.0 | 4.4                   | 80   | 0.519 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 16.0                   | 83.7            | 66.5 | 80.6 | 65.4 | 77.5 | 64.2 | 68.4                | 79.9 | 67.0 | 77.8 | 65.4 | 75.1 | 6.2                  | 180  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 64.5                            | 93.0 | 72.4 | 62.9 | 88.1 | 71.1 | 61.4 | 83.4 | 69.9 | 33.0          | 80.9 | 31.8 | 78.2 | 30.6 | 75.4 | 75.9              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |     |     |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|-----|-----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |     |     |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB  | WB  |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |     |     |
| (4) 23.2          | 20.3 | 17.7 | 13.6                       | 91.0 | 8.4                | 3.1 | 7.6                                                | 93.2 | 2.7        | 95.0 | -2.0       | 96.8 | -8.1       | 99.0 |     |     |
| (5)               |      |      | 12.4                       | 71.5 | 8.1                | 2.1 | 6.6                                                | 73.0 | 1.9        | 74.3 | -2.6       | 75.5 | -8.5       | 77.0 | (4) | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 50.4        | 36.2  | 36.0  | 44.0  | 47.2  | 55.2  | 60.6  | 65.2  | 64.8  | 58.5  | 52.1  | 44.3  | 40.2  | (6)  | (7)  |
|                                                                        | DBStd       | 12.23       | 8.99  | 9.65  | 6.20  | 6.68  | 6.89  | 6.47  | 5.98  | 5.54  | 5.48  | 6.59  | 7.23  | 6.43  | (7)  | (8)  |
|                                                                        | HDD50       | 1754        | 428   | 396   | 208   | 131   | 27    | 1     | 0     | 0     | 2     | 53    | 198   | 310   | (8)  | (9)  |
|                                                                        | HDD65       | 5520        | 892   | 811   | 652   | 535   | 313   | 166   | 75    | 73    | 207   | 404   | 622   | 770   | (9)  | (10) |
|                                                                        | CDD50       | 1916        | 1     | 5     | 21    | 47    | 187   | 319   | 472   | 458   | 256   | 118   | 26    | 6     | (10) | (11) |
|                                                                        | CDD65       | 205         | 0     | 0     | 0     | 0     | 8     | 34    | 82    | 65    | 12    | 4     | 0     | 0     | (11) | (12) |
|                                                                        | CDH74       | 1518        | 0     | 0     | 0     | 0     | 84    | 233   | 652   | 459   | 83    | 7     | 0     | 0     | (12) | (13) |
|                                                                        | CDH80       | 324         | 0     | 0     | 0     | 0     | 10    | 24    | 159   | 122   | 9     | 0     | 0     | 0     | (13) | (14) |
| (14) Wind                                                              | WSAvg       | 7.0         | 9.5   | 8.0   | 8.4   | 7.4   | 5.6   | 5.5   | 4.9   | 5.4   | 6.2   | 6.7   | 8.0   | 8.9   | (14) | (15) |
| (15) Precipitation                                                     | PrecAvg     | 32.0        | 2.7   | 2.6   | 2.2   | 1.9   | 2.4   | 3.0   | 3.1   | 3.3   | 2.5   | 2.5   | 2.7   | 3.1   | (15) | (16) |
|                                                                        | PrecMax     | 38.9        | 5.4   | 6.0   | 4.5   | 3.8   | 5.1   | 6.3   | 7.3   | 6.7   | 6.2   | 6.0   | 4.5   | 6.9   | (16) | (17) |
|                                                                        | PrecMin     | 25.7        | 0.2   | 0.4   | 0.2   | 0.0   | 0.9   | 0.7   | 1.2   | 1.0   | 0.6   | 1.3   | 0.2   | 0.9   | (17) | (18) |
|                                                                        | PrecStd     | 3.7         | 1.3   | 1.3   | 1.1   | 0.9   | 1.0   | 1.3   | 1.6   | 1.7   | 1.5   | 1.0   | 1.1   | 1.3   | (18) | (19) |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 53.7  | 61.0  | 67.2  | 70.6  | 80.8  | 82.9  | 88.1  | 89.1  | 81.2  | 74.9  | 62.9  | 58.2  | (19) | (20) |
|                                                                        |             | MCWB        | 48.3  | 51.0  | 52.2  | 54.0  | 61.4  | 66.7  | 68.0  | 68.0  | 63.9  | 63.0  | 53.4  | 49.8  | (20) | (21) |
|                                                                        | 2%          | DB          | 51.1  | 55.4  | 61.4  | 66.3  | 76.2  | 79.7  | 84.2  | 83.1  | 76.1  | 68.8  | 59.1  | 55.3  | (21) | (22) |
|                                                                        |             | MCWB        | 46.2  | 48.3  | 50.3  | 51.9  | 60.1  | 65.3  | 67.2  | 66.4  | 63.3  | 60.1  | 51.9  | 49.4  | (22) | (23) |
|                                                                        | 5%          | DB          | 48.9  | 52.1  | 56.6  | 62.0  | 71.8  | 76.8  | 81.1  | 79.0  | 71.3  | 64.9  | 56.4  | 51.7  | (23) | (24) |
|                                                                        |             | MCWB        | 45.7  | 46.7  | 49.5  | 50.5  | 57.9  | 64.0  | 66.2  | 65.1  | 60.8  | 57.9  | 51.3  | 47.7  | (24) | (25) |
|                                                                        | 10%         | DB          | 46.7  | 48.8  | 53.4  | 58.4  | 67.8  | 72.7  | 77.6  | 75.1  | 67.5  | 61.4  | 54.0  | 48.8  | (25) | (26) |
|                                                                        |             | MCWB        | 43.7  | 44.8  | 47.7  | 48.9  | 56.6  | 62.0  | 65.1  | 63.8  | 59.2  | 55.8  | 50.2  | 45.6  | (26) | (27) |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 50.6  | 52.5  | 53.7  | 56.2  | 63.6  | 69.2  | 70.8  | 70.2  | 66.2  | 63.4  | 55.5  | 52.1  | (27) | (28) |
|                                                                        |             | MCDB        | 52.9  | 58.4  | 63.4  | 66.4  | 77.0  | 78.6  | 82.4  | 81.8  | 78.1  | 71.9  | 59.0  | 55.3  | (28) | (29) |
|                                                                        | 2%          | WB          | 47.8  | 49.7  | 51.9  | 53.3  | 61.3  | 67.1  | 68.9  | 67.8  | 63.8  | 61.1  | 53.5  | 50.2  | (29) | (30) |
|                                                                        |             | MCDB        | 49.6  | 54.4  | 58.8  | 62.7  | 72.8  | 76.9  | 80.8  | 79.6  | 72.6  | 67.5  | 56.9  | 54.4  | (30) | (31) |
|                                                                        | 5%          | WB          | 46.0  | 47.1  | 50.1  | 51.5  | 59.2  | 65.3  | 67.4  | 66.3  | 62.0  | 58.8  | 52.0  | 48.1  | (31) | (32) |
|                                                                        |             | MCDB        | 48.2  | 51.3  | 56.3  | 60.7  | 69.2  | 74.4  | 78.7  | 76.7  | 68.8  | 64.1  | 55.9  | 51.1  | (32) | (33) |
|                                                                        | 10%         | WB          | 44.0  | 44.8  | 48.0  | 49.6  | 57.2  | 63.0  | 65.6  | 64.7  | 60.5  | 56.1  | 50.1  | 46.2  | (33) | (34) |
|                                                                        |             | MCDB        | 46.7  | 48.6  | 52.6  | 57.4  | 66.3  | 70.8  | 75.6  | 73.5  | 66.5  | 60.3  | 53.8  | 48.6  | (34) | (35) |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 7.7   | 10.5  | 11.4  | 13.8  | 14.8  | 14.6  | 16.0  | 15.3  | 13.5  | 10.9  | 8.7   | 7.2   | (35) | (36) |
|                                                                        |             | MCDBR       | 10.1  | 14.1  | 16.5  | 20.1  | 22.3  | 20.9  | 22.0  | 21.6  | 20.0  | 14.8  | 11.8  | 10.3  | (36) | (37) |
|                                                                        | 5% WB       | MCWBR       | 8.3   | 9.6   | 9.7   | 10.4  | 11.0  | 10.5  | 10.0  | 9.5   | 9.9   | 8.4   | 8.1   | 7.8   | (37) | (38) |
|                                                                        |             | MCWBR       | 8.6   | 12.8  | 13.9  | 17.7  | 19.9  | 19.1  | 20.0  | 18.3  | 16.8  | 13.1  | 10.5  | 9.2   | (38) | (39) |
| (39) Clear-Sky Solar Irradiance                                        | taub        | taub        | 0.314 | 0.337 | 0.369 | 0.403 | 0.410 | 0.412 | 0.415 | 0.407 | 0.378 | 0.354 | 0.330 | 0.310 | (39) | (40) |
|                                                                        |             | taud        | 2.429 | 2.379 | 2.306 | 2.236 | 2.255 | 2.266 | 2.256 | 2.303 | 2.370 | 2.433 | 2.429 | 2.402 | (40) | (41) |
|                                                                        | Ebn at Noon | Ebn at Noon | 227   | 251   | 262   | 265   | 268   | 268   | 265   | 261   | 256   | 241   | 218   | 208   | (41) | (42) |
|                                                                        |             | Edh at Noon | 19    | 26    | 33    | 40    | 41    | 41    | 41    | 37    | 31    | 24    | 19    | 17    | (42) | (43) |
| (43) All-Sky Solar Radiation                                           | RadAvg      | RadAvg      | 241   | 457   | 818   | 1305  | 1549  | 1691  | 1602  | 1370  | 1019  | 598   | 281   | 186   | (43) | (44) |
|                                                                        | RadStd      | RadStd      | 36    | 88    | 121   | 150   | 146   | 154   | 192   | 135   | 117   | 55    | 35    | 27    | (44) | (45) |

## Historical Trends

|                              |  | DBAvg | Heating |        |       | Cooling |       |     | Degree-Days |       |       |       |
|------------------------------|--|-------|---------|--------|-------|---------|-------|-----|-------------|-------|-------|-------|
|                              |  |       | 99% DB  | 99% DP | 1% DB | 1% WB   | 1% DP |     | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |  | (d)   | (e)     | (f)    | (g)   | (h)     | (i)   | (j) | (k)         | (l)   | (m)   |       |
| (46) Station Only            |  | N/A   | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | N/A   | N/A   | (46)  |
| (47) Regional (84 neighbors) |  | N/A   | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | N/A   | N/A   | (47)  |

Nomenclature: See separate page